Toward a four-dimensional Yang–Mills construction

Notebook 39: normalized local replacement in the bare Wilson measure

Research proof draft

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Abstract

We complete a normalized replacement operation for arbitrary finite collections of separated fixed boxes in the positive bare Wilson measure. On a good boundary the exact conditional law is restricted to its local gauge/minimizer window and divided by its actual conditional probability. On a bad boundary the law is unchanged. The operation preserves the exterior marginal, gauge invariance, and total raw partition mass. A positive expansion accounts for every discarded interior configuration.

An explicit coupling has discrepancy marks dominated by independent Bernoulli variables of parameter $a = Ce^{-cT^2}$. For the exterior jointly with any selected set A of box interiors, the exact total variation distance is $\mathbb{E}[1 - \prod_{i\in A}(1-p_i)] \le a|A|$. We also prove local entropy, sparse-defect, cluster, bounded-source and connected-cumulant comparison bounds. All are independent of the number of other boxes.

The probability inputs are the full normalized Wilson bounds of Notebook 38, not Gaussian surrogates. Geometry and box size remain fixed. The operation has not been identified with a nonlinear coarse-average fiber or propagated through general mixed renormalization histories. This completes a restricted finite-measure theorem and gives a definite stopping point for handoff; it does not complete continuum Yang–Mills existence.

Inputs and proof boundary

The two quantitative inputs are the fixed-box conditional tail theorem and joint boundary-plaquette probability theorem proved in Notebook 38. Both concern the actual compact positive Wilson measure and include their normalization. Their box geometry, boundary gauge, adjoint-invariant window, period conditions and constants are restated below. The new argument is an exact normalized operation built from those inputs, together with its local comparison estimates. It does not use the earlier homogeneous determinant calculation.

The group may be $SU(N)$ for these bare-measure statements, with $\tau = \mathrm{Tr}\,/N$. This does not extend the earlier $SU(2)$ multiscale construction to every group. An internal audit is included in the research archive. Such review and successful typesetting do not certify all upstream arguments independently.

1 A normalized fixed-box replacement for the full Wilson measure

This section proves a restricted normalized replacement theorem for the *actual positive bare Wilson measure*. It preserves the exterior marginal and the raw partition mass exactly, leaves bad-boundary boxes unchanged, and gives a sparse failure expansion and local comparison bounds uniform in the number of boxes. The fixed-box coordinates are not identified with a nonlinear coarse-average fiber.

1.1 Exact conditional structure and the established box estimates

Let

$$\mu(dU) = Z_W^{-1} e^{-S_W(U)/g^2} \prod_e dU_e$$

be the normalized positive Wilson measure on a finite compatible four-dimensional periodic lattice, with normalized link Haar measure. Fix $R \geq 2$, $b \geq 3$, and a finite collection I of embedded translates Ω_i of $[0, R]^4$, with pairwise disjoint vertex sets and no local periodic identifications inside a box. The surrounding periods are divisible by $2b$. Constants may depend on this fixed geometry and group, but not on the number or placement of boxes or on the surrounding volume.

Let $E_{\circ,i}$ be the edges of box i not contained in its topological boundary. The exterior variable ω comprises every edge outside their union, including all box boundary edges; let μ_{ext} be its actual marginal. Write u_i for the interior edges. Every plaquette meeting $E_{\circ,i}$ lies entirely in Ω_i. No plaquette therefore depends on two integrated interiors. The exact conditional law is

$$\mu(d\omega, du) = \mu_{\text{ext}}(d\omega) \prod_{i \in I} Q_i^{\omega}(du_i), \tag{1}$$

where Q_i^{ω} is the normalized compact Wilson law of that box with its actual boundary fixed. It depends only on that boundary. All exterior action terms cancel from this normalization.

Use the boundary-only tree gauge and then the forest rooted at the boundary vertices from the fixed-box result. Let $B_i(\omega)$ be the event that some boundary plaquette has $h(U_p) \geq \varepsilon_R^2$. On B_i^c, the transformed boundary lies in the fixed small-boundary neighborhood. Let $W_i(\omega)$ be the lifted window about its unique global conditional minimizer,

$$\|x - x_*(V_i(\omega))\|_\infty \leq gT, \tag{2}$$

using the invariant edge/color norm in the forest coordinates. It is an actual measurable gauge-invariant event: the full tree representative and its unique minimizer transform by the same constant root conjugation.

The already proved fixed-box conditional theorem supplies constants $C, c, r, g_0 > 0$ such that

$$Q_i^{\omega}(W_i^c) \leq a := Ce^{-cT^2} \quad \text{on } B_i^c, \qquad 0 < g \leq g_0, \quad 1 \leq T \leq r/g. \tag{3}$$

Choose a fixed $T_0 \geq 1$ so that $Ce^{-cT_0^2} \leq 1/2$, and henceforth take $T_0 \leq T \leq r/g$. Thus $a \leq 1/2$. This estimate uses the entire compact conditional integral: its $g^n e^{-S_{\min}/g^2}$ denominator and local numerator powers cancel, and its complementary compact region has a positive action gap. It is not a Gaussian-law substitution.

The full Wilson boundary estimate supplies, for every $S \subset I$,

$$\mu_{\text{ext}}\left(\bigcap_{i \in S} B_i\right) \leq \beta^{|S|}, \qquad \beta = \min\{1, 24R^3 z_b(g, \varepsilon_R)^{1/Q_b}\}, \quad Q_b = 6b^4, \tag{4}$$

where

$$z_b(g, \delta) = \min\left\{1, \exp\left[-\delta^2/g^2 + b^4\{4 \dim G \log(1/g) + C_{G,4}\}\right]\right\}.$$

Indeed the boundary events are exterior measurable. A union bound over one offending boundary plaquette in each box and the prescribed- plaquette Wilson estimate prove (4); the chosen plaquettes are distinct by vertex disjointness. Boundary independence is not assumed. These are fixed-geometry inputs from the full-measure box theorem, not assertions for arbitrary effective actions.

1.2 The normalized operation

Define

$$p_i(\omega) = \mathbf{1}_{B_i^c} Q_i^\omega(W_i^c), \qquad P_i^\omega = \begin{cases} Q_i^\omega(\cdot \mid W_i), & B_i^c, \\ Q_i^\omega, & B_i. \end{cases} \tag{5}$$

On good boundaries, $1 - p_i \geq 1 - a \geq 1/2$; on bad boundaries $p_i = 0$ and the conditional law is left unchanged. All kernels are measurable finite-dimensional compact integrals. For $J \subset I$, put

$$\nu_J(d\omega, du) = \mu_{\text{ext}}(d\omega) \prod_{i \in J} P_i^\omega(du_i) \prod_{i \notin J} Q_i^\omega(du_i), \qquad \nu = \nu_I. \tag{6}$$

Theorem 1.1 (Normalized fixed-box replacement)**.** *The measures ν_J are probability measures with exactly the original exterior marginal and with the Radon–Nikodym derivative*

$$\frac{d\nu_J}{d\mu} = \prod_{\substack{i \in J \\ B_i^c}} \frac{\mathbf{1}_{W_i}(u_i)}{1 - p_i(\omega)}. \tag{7}$$

They are gauge invariant. The same replacement applied to the unnormalized Wilson measure preserves its total mass Z_W exactly. There is a coupling of μ and ν with the same exterior whose box discrepancy marks are dominated by independent Bernoulli variables of parameter a. Bad-boundary interiors are identical in this coupling. In particular every fixed local observable admits comparison bounds independent of $|I|$, as quantified below.

Proof. Conditioning on ω and multiplying the normalized kernels proves (7). Each factor has conditional Q_i^ω expectation one. Thus the product has expectation one and

$$\int e^{-S_W/g^2} \prod_{\substack{i \in J \\ B_i^c}} \frac{\mathbf{1}_{W_i}}{1 - p_i} \prod_e dU_e = Z_W. \tag{8}$$

Every exterior observable is preserved exactly. Gauge covariance of conditional Haar/Wilson laws, invariance of B_i, and invariance of the lifted tree-gauge window show that (7) is gauge invariant. The explicit coupling is constructed in the next subsection. □

The conditional denominators are exact boundary-dependent quantities, not Gaussian determinants or vacuum constants. They obey

$$0 \leq -\log(1 - p_i) \leq \frac{p_i}{1 - a} \leq \frac{a}{1 - a}. \tag{9}$$

Their dependence is confined to the fixed box boundary. The full product density may nevertheless grow exponentially with $|J|$; no uniform global density-ratio bound is asserted. The individual kernels also define resampling operators: keep the exterior and redraw interior i from P_i^ω. These operators commute for distinct boxes and are idempotent, because no resampling changes another box's boundary. Their product applied once to μ gives ν.

1.3 An exact positive failure expansion and coupling

On a good boundary with $p_i > 0$, let $F_i^\omega = Q_i^\omega(\cdot \mid W_i^c)$. If $p_i = 0$, choose any probability kernel for F_i; it will be multiplied by zero. For every ω,

$$Q_i^\omega = (1 - p_i) P_i^\omega + p_i F_i^\omega. \tag{10}$$

Consequently

$$\mu = \sum_{S \subset I} \mathcal{M}_S,$$
$$\mathcal{M}_S(d\omega, du) = \mu_{\text{ext}}(d\omega) \prod_{i \in S} p_i \prod_{i \notin S} (1 - p_i) \prod_{i \in S} F_i^{\omega}(du_i) \prod_{i \notin S} P_i^{\omega}(du_i). \tag{11}$$

All terms are positive and their masses sum to one. The empty term is $\prod_i (1 - p_i(\omega))\, \nu$, not in general a constant multiple of ν. Thus no exterior reweighting is hidden in this expansion.

To couple the measures, sample $\omega \sim \mu_{\text{ext}}$. Conditionally and independently across boxes, sample $Y_i \sim P_i^{\omega}$, an independent Bernoulli mark D_i with parameter p_i, and, when needed, $Z_i \sim F_i^{\omega}$. Set

$$X_i = \begin{cases} Y_i, & D_i = 0, \\ Z_i, & D_i = 1. \end{cases} \tag{12}$$

Then $(\omega, X) \sim \mu$ and $(\omega, Y) \sim \nu$. On a marked box, $X_i \notin W_i$ and $Y_i \in W_i$; elsewhere they are identical. Therefore D_i is precisely its discrepancy indicator. On bad boundaries $p_i = 0$, so the interior remains the same sample from the original law.

Take independent uniform random variables U_i, independent of the exterior, and realize $D_i = \mathbf{1}_{U_i \le p_i(\omega)}$. Then $D_i \le \mathbf{1}_{U_i \le a}$ simultaneously. This proves the iid Bernoulli domination asserted in Theorem 1.1, and

$$\mathbb{P}(D_i = 1 \text{ for all } i \in S) = \mathbb{E} \prod_{i \in S} p_i \le a^{|S|}. \tag{13}$$

This stochastic domination concerns the coupling marks, not the possibly correlated bad boundaries.

1.4 Local total variation, observables and entropy

Use the convention $\|\lambda - \lambda'\|_{\text{TV}} = \sup_E |\lambda(E) - \lambda'(E)|$. For $A \subset I$, retain the whole exterior but only the interiors in A. Conditionally, their replacement law is their original law conditioned on all its good-boundary windows, of probability $\prod_{i \in A}(1 - p_i)$. Hence

$$\|\mu_{(\text{ext},A)} - \nu_{(\text{ext},A)}\|_{\text{TV}} = \mathbb{E}_{\mu_{\text{ext}}} \left[1 - \prod_{i \in A} (1 - p_i) \right] \le a|A|. \tag{14}$$

The equality follows by integrating the conditional positive and negative density parts; equivalently, the coupling is identical with that conditional probability, and any discrepancy puts the original field outside the replacement support. For the marginal on A alone the same expression is an upper bound.

For a bounded observable supported on the exterior and these interiors,

$$|\mu(f) - \nu(f)| \le 2\|f\|_{\infty} a|A|. \tag{15}$$

More generally, let $\text{osc}_i(f)$ be the supremum of the change when only box i's interior changes. Telescoping along the coupling proves

$$|\mu(f) - \nu(f)| \le \sum_i \text{osc}_i(f)\, \mathbb{E} p_i \le a \sum_i \text{osc}_i(f). \tag{16}$$

Thus local error depends on the region read by the observable, not on the total number of replaced boxes. Global total variation can approach one as that number grows.

The local joint-with-exterior relative entropy is also exact:

$$D(\nu_{(\text{ext},A)} \| \mu_{(\text{ext},A)}) = \mathbb{E}_{\mu_{\text{ext}}} \sum_{i \in A} -\log(1 - p_i) \le \frac{|A|a}{1 - a}. \tag{17}$$

Substitute the local version of (7); all its window indicators equal one under ν. No assertion about finiteness of the reverse entropy is needed.

1.5 Untreated boundaries, exponential moments and clusters

The exterior marginal is unchanged, so

$$\nu\left(\bigcap_{i\in S} B_i\right) = \mu\left(\bigcap_{i\in S} B_i\right) \le \beta^{|S|}. \tag{18}$$

Under ν, the original box defect (bad boundary or a failed window on a good boundary) is exactly B_i almost surely. Its good-boundary part has been removed by an exactly normalized conditional operation; its bad-boundary law has not been discarded or altered.

On the coupling put $C_i = \mathbf{1}_{B_i} \vee D_i$. These alternatives are disjoint at one box. Expanding a prescribed intersection over the subset assigned to the boundary alternative, and conditioning on the exterior, gives

$$\mathbb{P}(C_i = 1 \text{ for all } i \in S) \le \sum_{J\subset S} a^{|S|-|J|}\mu_{\text{ext}}\left(\bigcap_{i\in J} B_i\right) \le (a+\beta)^{|S|}. \tag{19}$$

Let $\varepsilon = \min(1, a+\beta)$. Expansion of the finite product proves, for $t \ge 0$,

$$\mathbb{E}\exp\left(t\sum_{i\in A} C_i\right) \le [1+\varepsilon(e^t-1)]^{|A|}. \tag{20}$$

The discrepancy version has a in place of ε, even conditionally on the exterior. This joint-inclusion argument does not assert iid stochastic domination for C. For $1 \le k \le |A|$, a union bound gives

$$\mathbb{P}\left(\sum_{i\in A} C_i \ge k\right) \le \min\left\{1, \binom{|A|}{k}\varepsilon^k\right\} \le \min\{1, (e|A|\varepsilon/k)^k\}. \tag{21}$$

For $k > |A|$ the probability is zero. One may also optimize the Chernoff estimate from (20).

On any fixed adjacency graph of boxes with maximum degree $\Delta \ge 2$, there are at most Δ^{2m} connected m-subsets containing a given vertex: encode a canonical spanning tree by its depth-first traversal. A cluster of size at least m contains such a subset. Therefore

$$\mathbb{P}(|C\text{-cluster at } i| \ge m) \le [\Delta^2\varepsilon]^m. \tag{22}$$

The same argument gives a for discrepancies and β for defects under the replacement measure. These estimates are uniform in volume and the total number of boxes, at the fixed geometry/degree. For $T(g) = (\log(1/g))^\alpha$, $\alpha > 1/2$, both a and β are smaller than every fixed power of g at small coupling. All fixed-region bounds inherit that property.

1.6 Exact source identities and local connected cumulants

For bounded measurable single-box sources $f_i(\omega, u_i)$ and an exterior source $f_{\text{ext}}(\omega)$, conditional independence gives

$$\nu\left(e^{f_{\text{ext}}+\sum_i f_i}\right) = \int \mu_{\text{ext}}(d\omega)e^{f_{\text{ext}}}\prod_i P_i^\omega(e^{f_i}), \tag{23}$$

$$\mu\left(e^{f_{\text{ext}}+\sum_i f_i}\right) = \int \mu_{\text{ext}}(d\omega)e^{f_{\text{ext}}}\prod_i [(1-p_i)P_i^\omega(e^{f_i}) + p_i F_i^\omega(e^{f_i})]. \tag{24}$$

On good boundaries, $P_i^\omega(e^{f_i}) = Q_i^\omega(\mathbf{1}_{W_i}e^{f_i})/(1-p_i)$; on bad boundaries it is unchanged. The exact zero-source conditional masses remain in these denominators. No exterior-dependent factor is absorbed

into a vacuum constant. The identities are positive expansions for real sources and absolutely valid for bounded complex sources. Cross-box sources use the full measure expansion, not a false product factorization in (23).

For any bounded F depending on the exterior and interiors in A, $\|F\|_\infty \le M$, the coupling yields

$$|\mu(e^F) - \nu(e^F)| \le 2e^M a|A|. \tag{25}$$

For real F, both expectations are at least e^{-M}, so their real logarithms differ by at most $2e^{2M}a|A|$.

Now let $O_1, \dots, O_q$ be bounded observables with the same fixed local dependence, put $M_l = \|O_l\|_\infty > 0$, and omit zero observables. Fix $0 < s_0 < \log 2$, $d_0 = 2 - e^{s_0} > 0$. For $F(t) = \sum_l t_l O_l$ and $Z_\lambda(t) = \lambda(e^{F(t)})$, the weighted source ball $\sum_l |t_l| M_l \le s_0$ satisfies

$$|Z_\mu(t) - 1|,\ |Z_\nu(t) - 1| \le e^{s_0} - 1 < 1. \tag{26}$$

Both functions are nonzero, with analytic logarithms zero at the origin on the same branch. The straight segment between their values also remains in this disk. Integrating $1/z$ along it and using (25) gives

$$|\log Z_\mu(t) - \log Z_\nu(t)| \le \frac{2e^{s_0}}{d_0}\, a|A|. \tag{27}$$

For another bounded observable O whose support is included in A, the normalized source numerator obeys

$$\left| \frac{\mu(Oe^F)}{Z_\mu} - \frac{\nu(Oe^F)}{Z_\nu} \right| \le \frac{4\|O\|_\infty e^{s_0}}{d_0^2}\, a|A|. \tag{28}$$

Indeed the numerator difference is at most $2\|O\|_\infty e^{s_0} a|A|$, the denominator difference at most $2e^{s_0}a|A|$, their moduli are at least d_0, and $|\nu(Oe^F)| \le \|O\|_\infty e^{s_0}$. Use $d_0 + e^{s_0} = 2$ to obtain the displayed constant.

Let $k = (k_1, \dots, k_q) \ne 0$, $n = \sum_l k_l$. Cauchy on radii $s_0/(qM_l)$ gives the explicit connected-cumulant comparison

$$\left| \partial^k (\log Z_\mu - \log Z_\nu)(0) \right| \le \frac{2e^{s_0}}{d_0}\, a|A|\, k! \left(\frac{q}{s_0} \right)^n \prod_l M_l^{k_l}. \tag{29}$$

Alternatively set inactive variables to zero and use radii $s_0 k_l/(nM_l)$. The factor preceding $\prod M_l^{k_l}$ then improves from $k!(q/s_0)^n$ to

$$\frac{k!(n/s_0)^n}{\prod_{k_l>0} k_l^{k_l}} \le n!(e/s_0)^n.$$

The last inequality uses $k! \le \prod k_l^{k_l}$ and $n^n \le e^n n!$. All finite generating functions are entire for bounded observables, so differentiation under their integrals is valid. Windows and coupling are held fixed during differentiation. These statements are for bounded observables with fixed local support; uniformity in a later cutoff applies only if their displayed bounds remain uniform. They do not constitute estimates for renormalized unbounded fields.

1.7 Natural stopping point and scope

This operation treats an arbitrary finite union of vertex-disjoint fixed boxes in the full positive bare measure. Its conditional normalizations are exact; it preserves the exterior marginal, total partition mass and gauge invariance; and it accounts for every removed configuration through a positive sparse expansion. Unlike conditioning the whole lattice on a simultaneous good event, it does not reweight the exterior distribution. It also supplies local comparison, source, cumulant and cluster bounds uniform in the number of boxes.

The restriction is substantive. The box size and forest-coordinate conditional minimizer are fixed. No identification with an averaged coarse-constraint fiber, arbitrary growing component, collared multiscale determining set, scale-age estimate or effective action after renormalization is claimed. Bad-boundary boxes remain with their original conditional laws. Complex source formulas here come from bounded finite observables, not a complex-measure stability theorem. The result is a normalized local replacement input for future mixed-history analysis; it does not close the continuum existence, reconstruction, nontriviality or mass-gap problem.